

CMPE-131: Project

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Scenario title: The Thrifty Backpacker

Context & Requirement

A college student planning a trip to London needs to find a flight, the cheapest hostel, and free walking tours—all within a strict total budget of \$1,500.

Key Features to implement

Budget-based Filtering (Aggregation & Cap Logic)

Section A: Persona Profile

- **Name & Archetype:** Violet the Budget Backpacker
- **Background:**
 - Age: 21
 - Occupation: College Student
 - Tech-savviness: Low patience, digital native

As a college student Violet is planning a trip to London knowing she has a tight budget. She has been working as a part time barista while saving up for her first solo international trip. It will be her first time booking an international trip and is feeling anxious about any hidden fees, scams, and ending spending more than her budget. She is usually on her phone figuring out any handful of apps that may help, but she tends to give up the moment she does more than a few taps.

- **Goal & Motivations:**
 - Visit London for a week and stay strictly within her \$1,500 total trip budget.
 - Look for the must-do parts of London without paying expensive prices.
 - Find the cheap accommodation but also safe, and clean.
 - Find out free walking tours and free activities to enjoy

- **Pain Points:**

- Sites usually don't express the end total for flights and hotels separately until the end
- In the sites there is usually filters labeled cheapest but it usually shows part of the cost not the full cost of the trip
- Finding free activities could be difficult to find, that's why using different websites can be beneficial
- There isn't any warning when the user over selects options over their budgets, which in results leads to overspending

Section B: User stories

US-1 - Total-budget filtering

As a student traveler, I want to enter my total trip budget once in the program and it shows the flight and accommodations that fit in my budget, so that I don't have to waste my time looking for the trip that I cannot afford.

US-2 - Free & Low-cost Activity Discovery

As a budget-stricted backpacker, I want to filter activities by price, so that I can build an affordable trip in London.

US-3 - Always - visible running trip total

As a low-patience user, I want a program that shows trip totals instantly, even if I modify a flight, accommodation, or activity, so that I always keep tracking whether I'm still under budget without doing math myself.

US-4 - Trustworthy, branded agency experience (Multi-tenant SaaS)

As a customer who reached this platform through a specific travel agency, I want to see that agency's information, so that I trust who I'm booking with.

Section C: Acceptance Criteria

US-1 : Total budget filtering

- AC 1.1: a whole dollar amount tends to land between \$1 to \$100,000 which provides a total search of total budget in USD.
- AC 1.2: Whenever the user is finished with their search the website only tends to show the flights and the hostels combined cost which can demonstrate if the cost is less or more than the user's original budget.
- AC 1.3: At the end the card results displays the breakdown of the cost of flights, hostel, and combined total in USD.
- AC 1.4: Whenever a combination doesn't fit inside the specific budget the system displays a "no trip found within this budget" instead of displaying an empty page or an error.
- AC 1.5: It rejects any negative numbers, zero, or non numerical input, and any values above the \$100,000 which later gives an inline validation message before API call is called.

US-2: Free and Low Cost Activity

- AC 2.1: Within the activity panel it sometimes offers a combination of price filters: free, under \$25, \$25-\$75, and \$75 or more.
- AC 2.1: Since the destination is London the system displays around three different types of activities as free, for instance, walking tours, museums, and much more.
- AC 2.3: Whenever the user selects an activity it adds the price to the trips total in real time to see the cost in total.
- AC 2.4: In the activity card it demonstrates a precise price label, even in the activities that are considered free.

US-3: Trip Total

- AC 3.1: At the top of the search, it will display the results of the trip total widget, and it doesn't make the user scroll to find their information.
- AC 3.2: The color of the total trip is going to be green less than equal to 90% of budget, yellow would be considered between 90% and 100%, while the color red will be when it's over the budget.
- AC 3.3: whenever there is a remove, swap, or add action being made the total update will update within a minute.
- AC 3.4: The book now button can be disabled and a tooltip can explain why when the running total tends to exceed the original budget.

US-4: Multi-tenant SaaS branding & isolation

- AC 4.1: The page header displays the logged-in tenant's logo, agency name, and primary brand color.
- AC 4.2: All booking records are stored with a tenant_id and are only returned to users belongin to that same tenant - under no condition can a user from Tenant A see a booking that belongs to Tenant B.
- AC 4.3: If the tenant has not configured a logo or color, the system falls back to a neutral default theme rather than crashing or showing a broken image.

Section D - Test Cases

TC-01 - Happy path - search within budget returns valid combos

Test ID: TC - 01

Title: Search within budget returns valid combos

Linked AC: AC 1.1, AC 1.2, AC 1.3

Type: Happy path

Pre-conditions: User is logged in as a customer of Tenant A. The Amadeus sandbox API is reachable.

Test Steps:

1. Navigate to the Search page.
2. Enter destination = LON, valid dates, travelers = 1, total budget = \$1,500.
3. Click Search.

Expected Result:

The results page loads in under 3 seconds and displays one to three flights & accommodations. Every displayed combination has flights & accommodations under \$1,500. Each card shows the flight cost, accommodation cost, and combined total.

TC-02 - Sad path - budget too small, no results

Test ID: TC-02

Title: Budget too small, no results

Linked AC: AC 1.4

Type: Sad Path

Pre-conditions: User is logged in as a customer of Tenant A.

Test Steps:

1. Open the search page.
2. Enter destination = LON, valid dates, travelers = 1, total budget = \$50
3. Click Search

Expected Result:

The system does not crash and does not show an empty page. It displays the message:
“No trips found within \$50.”

TC-03 - Sad path - budget of \$0 is rejected before API call

Test ID: TC-03

Title: budget of \$0 is rejected before API call

Linked AC: AC 1.5

Type: Sad Path/ Edge Case

Pre-conditions: The user is logged in as a customer of Tenant A.

Test Steps:

1. Open the search page
2. Enter the budget = 0
3. Click search

Expect result: An inline validation message is displayed and it shows no request is sent to the Amadeus API (message displayed is “Please enter a budget greater than \$0”)

TC-04 - Free activity filter returns only \$0 activities

Test ID: TC-04

Title: Free activity filter returns only \$0 activities

Linked AC: AC 2.1, AC 2.2, AC 2.4

Type: Happy path

Pre-conditions: User is on the results page for a London search

Test Steps:

1. Open the activities panel
2. Enable only the “Free” filter

Expected Result:

Every listed activity has price = \$0 and shows the green “Free” label. At least five activities are returned.

TC-05 - Running Total Trip Updates within a second

Test ID: TC-05

Title: Running updates of the total trip within a second

Linked AC: AC 3.1 and AC 3.3

Type: Happy path and performance

Pre-conditions: Being in the result page with a \$1,500 budget

Test Steps:

1. Add hostel A (\$30/night * 5 nights)

2. Swap the hotel to B (25/night * 5 nights)

Expected Result: The total trip widget decreases by \$25 within a second of the swap action being made. Which means that the widget will stay visible without the user scrolling to find it.

TC-06 - Over-budget combination disables Book Now

Test ID: TC - 06

Title: Over-budget combination disables Book Now

Linked AC: AC 3.2, AC 3.4

Type: Sad Path

Pre-conditions: User has a \$1,500 budget and has selected items totaling \$1,700.

Test Steps:

1. Try to click the "Book Now" button

Expected Result:

The Trip Total widget is red. "Book Now" button is disabled. Hovering it shows a tooltip explaining the user is \$200 over budget.

TC-07 - Multi-tenant isolation - Tenant A booking is invisible to Tenant B

Test ID: TC - 07

Title: Multi-tenant isolation - Tenant A booking is invisible to Tenant B

Type: Security / SaaS

Pre-conditions: User1 belongs to Tenant A, User2 belongs to Tenant B. Both accounts exist

Test Steps:

1. Log in as User1 and create a booking to London
2. Log out
3. Log in as User2
4. Navigate to the Booking dashboard
5. Also, attempt to access User1's booking by guessing its ID in the URL

Expected Result:

User2's dashboard contains zero bookings. The direct-URL attempt returns a 403/404 - never User1's data

TC - 08 - Multi - tenant branding

Test ID: TC-08

Title: Each tenant tends to view their own theme

Linked AC: AC 4.1 and AC 4.3

Type: SaaS / UI

Pre-Conditions: The tenant A is required to configure with the blue theme and logo. While tenant B is required to be the green theme and logo. Tenant C doesn't have a theme configured.

Test Steps:

1. Login as tenant A and observe the header
2. Log out
3. Login as tenant B and observe the header
4. Log out
5. Login as tenant C and observe the header

Expected Result: Tenant A is supposed to see the blue logo and theme, tenant B is supposed to see the green theme and logo, and lastly tenant C is supposed to see no of them at all. Therefore, none of the other tenants tend to see each other's tenant's branding.